

TrueYield recovery report — Fitpack Textiles rooftop, Jaipur (system JAIPUR1)

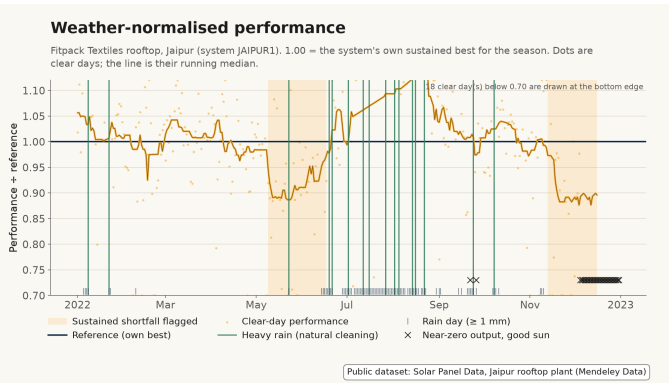
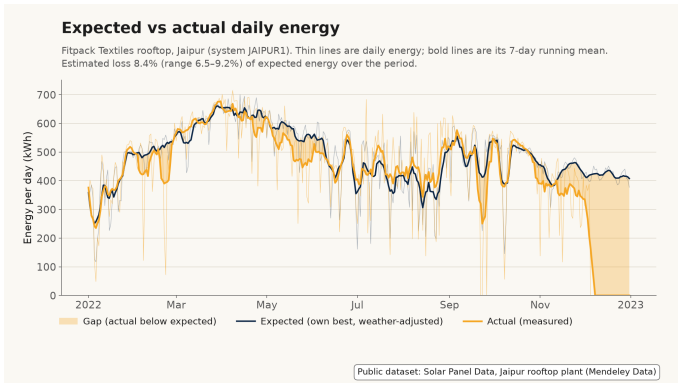
Data label: Public dataset: Solar Panel Data, Jaipur rooftop plant (Mendeley Data) | Period 2022-01-01 to 2022-12-31 | 120 kWp, tilt and azimuth not provided by the source | generated 2026-10-04

1. What is measured (facts)

- Generation: 365 valid days of 365 (100.0% complete); 0 days with no data, 0 partial days and 0 days with impossible values were excluded, not filled in.
- Loader inferred: JAIPUR1_daily.csv: system JAIPUR1; time column 'Date'; interval_energy from ['Energy (kWh)'] in kWh; 1440-minute steps. system_id taken from the file name.
- Weather: NASA POWER satellite irradiance (all-sky and clear-sky), 2 m temperature and rainfall for the site coordinates. No on-site sensor is used.

2. What is estimated, and how sure

- Estimated loss: 8.4% of expected energy (range 6.5–9.2%), 14,894 kWh (range 11,232–16,366 kWh) over 365 valid days. Of this, 2,743 kWh (range 0–4,399) is sustained shortfall in 2 flagged episode(s), and 12,151 kWh (range 10,787–12,454) is from days with near-zero output. Ranges are 95% bootstrap intervals covering day-to-day noise and the uncertainty of the reference. Shortfalls below the trigger are counted as zero, so this is a conservative figure.
- Detection limit: losses below about 9.1% sustained over 14 days, or about 12.0% sustained over 30 days, cannot be distinguished from noise (from clear-day scatter of 6.7% per day; 1% false-alarm rate per window, 80% power, starting from a system at its reference). Checked with injected-loss trials on real data from system JAIPUR1: reliable detection (at least 80% of trials) was reached not at any size tried over 14 days, and at about 24% over 30 days. Smaller losses are counted as zero.
- Method: Daily mode. Daily horizontal irradiation × nameplate capacity × a temperature factor (−0.4%/°C, NOCT 45 °C cell-temperature rule). Performance index = measured ÷ modelled energy. The reference is the system's own 90th-percentile sustained (7-clear-day median) performance per calendar month, pooled over all years and reduced by the upward bias expected from noise. Clear day = all-sky ÷ clear-sky irradiation ≥ 0.85. Assumptions: constant inverter and wiring losses are absorbed by the reference; timestamps are local standard time; the cleanest state seen in each month is representative of a clean array.



3. Likely causes: detected facts vs hypotheses

Cause	Detected facts	Hypothesis (not proven)
outage or fault indicated	Output was near zero on 29 day(s) with good irradiance: 2022-09-22; 2022-09-26; 2022-12-05 to 2022-12-31 (27 days).	Near-zero days are consistent with an inverter trip, grid outage or a monitoring fault that recorded zeros. The data cannot tell these apart.
soiling not indicated	Between cleaning events, performance declined in 2 of 2 usable intervals, at a median of 0.232% per day in the declining ones. 1 of 7 testable natural rain event(s) were followed by a rise in normalised performance that is distinguishable from zero. rdttools estimates a soiling loss of 5.3% (range 1.3–12.1%, rdttools 3.2.1 stochastic rate and recovery (SRR)). Shuffle check: with the same clear-day values in random time order, which contain no soiling pattern, it reports 6.0–12.4%.	No soiling distinguishable from scatter. The estimate of 5.3% (range 1.3–12.1%, rdttools 3.2.1 stochastic rate and recovery (SRR)) is not clearly above what the same method recovered on shuffled data (up to 12.4%), so most or all of it can be produced by day-to-day scatter alone.
ageing or degradation insufficient data	Only 1.0 years of data; more than 2 years are needed to separate ageing from seasonal effects.	—
shading cannot be assessed	Cannot be assessed from daily data: shading shows as dips at fixed hours.	—

4. Recovery verification

- Events tested: natural rain event, not a maintenance intervention. 7 of 16 event(s) had enough clear days on both sides (14 days before, 14 after).
- Best-observed event, 2022-05-23 (natural rain event, not a maintenance intervention): +0.6% (range −12.9 to +17.2%), +2.9 kWh/day (range −67.6 to +77.1), from 14 clear days before and 13 after. It was chosen for having the most clear days, not for its size. The range includes zero: no recovery can be claimed for this event.
- Across all testable events: 1 clear rise(s), 0 clear fall(s), 6 not distinguishable from zero. Full list in summary.json.

5. Value and verdict

- [not available: tariff_inr_per_kwh not provided in data/systems.csv]
- Emissions: [not available: emission_factor_kg_per_kwh is empty in config.yaml; no factor is assumed]

6. Limits of this report

- Causes are hypotheses; none is proven without a site inspection.
- Loss is measured against this system's own best month by month; a loss present from the start, or one that recurs in the same month every year, is partly or wholly invisible.
- Irradiance is a satellite area average; losses below the detection limit are not counted.
- Tilt and azimuth are not provided by the data source. Daily mode does not use them (the seasonal reference absorbs the orientation), so no value was assumed.
- Commissioning date is not provided by the data source; it is not used in any calculation.
- Months with too few clear days for their own reference (borrowed from neighbouring months): [7, 8, 12].

7. Method validation with injected loss — not a field result

- On a clean real stretch of this system (2022-01-01 to 2022-08-28), a 10% soiling ramp and a 3-day outage were injected (3,529 kWh). The unchanged pipeline recovered 1,411 kWh (40% of the injected loss; estimate 3,263 kWh, range 1,245–4,276); the known truth is OUTSIDE the stated range. Outage days found: 3 of 3. Detection-limit check: over 14 days: 5% step found in 10%, 9% step found in 17%, 14% step found in 30%, 18% step found in 73%, 23% step found in 77%, 27% step found in 77%; over 30 days: 6% step found in 20%, 12% step found in 43%, 18% step found in 77%, 24% step found in 80%, 30% step found in 90%, 36% step found in 90% of 30 trials each. These are test results on altered data, not field results. The method did not pass this check on this dataset: treat its sustained-shortfall figures as rough, and its stated range as too narrow.